An exact compress-with-another threshold in a binary automaton family

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Abstract

For a state q in a synchronizing automaton, let $\mu(q)$ be the minimum length of a word merging q with another initial state. Szykuła asked whether $\mu(q) = O(n)$ for strongly connected synchronizing automata and, following Dżyga, proposed a binary family on $n = 3p$ states with threshold $4n/3$. We specify the boundary transitions omitted from the published piecewise definition and prove that the resulting automata $\mathcal{A}_p$ satisfy $\mu(q_0) = 4p = 4n/3$ for all $p \geq 3$. A shortest merging word is $(ba)^p(ab)^p$. We also construct a reset word of length $3p^2 + 4p - 1$. Consequently, any uniform linear upper bound for $\mu(q)$ on binary strongly connected synchronizing automata with corank-one letters has leading constant at least $4/3$.

Keywords. Synchronizing automata; compress-with-another threshold; pair automata; reset words.
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1 Introduction

A complete deterministic semiautomaton $\mathcal{A} = (Q, \Sigma, \delta)$ is *synchronizing* if some word maps Q to a singleton. The minimum length of such a *reset word* is the *reset threshold* $\mathrm{rt}(\mathcal{A})$. With transitions acting on the right, $qw = \delta(q, w)$, the *compress-with-another threshold* of $q \in Q$ is

$$\mu(q) = \min\{|w| : qw = rw \text{ for some } r \in Q \setminus \{q\}\}.$$

Unlike the reset threshold, $\mu(q)$ concerns a collision involving one prescribed initial state. The problem is related to the Černý conjecture on reset words [1, 2].

Every pair of states in a synchronizing automaton can be merged. Removing cycles from a shortest path in the distinct-pair graph gives the elementary general bound $\mu(q) \leq \binom{n}{2}$. Szykuła's Conjecture 6 [4] asks whether $\mu(q) = O(n)$ holds uniformly for *strongly connected* synchronizing automata. Strong connectivity matters: Proposition 7 of [4] constructs synchronizing automata with $\mu(q) = \Theta(n^2)$ without it. Figure 3 and Conjecture 8 of the same survey, attributed to Dżyga [3], propose a binary family on $3k + 6$ states conjectured to attain $\mu(q_0) = 4n/3$.

Here $p = k + 2 \geq 3$ and $n = 3p$. For the completed family $\mathcal{A}_p$ defined in Section 2, we prove

$$\mu(q_0) = 4p, \qquad \mathrm{rt}(\mathcal{A}_p) \leq 3p^2 + 4p - 1.$$

The first equality gives a lower bound of $4/3$ on the coefficient of any uniform linear upper bound; the existence of such a bound remains open. The piecewise definition in [4] omits $q_0 b$ and one terminal transition, depending on the parity of p; we use the edges drawn in its Figure 3 to complete them.

The collision lower bound uses pair-graph entrance estimates, and the synchronization proof constructs a reset word separately. Other publicly available accounts of this family appear in [5, 6].

2 The automaton family $\mathcal{A}_p$

Let $\mathcal{A}_p$ have $Q = \{q_0, \dots, q_{3p-1}\}$ and alphabet $\{a, b\}$. The transitions are specified by matchings and a few exceptional edges. Letter a fixes q_0, swaps q_{2j-1} and q_{2j} for $1 \le j \le p-1$, and has

$$q_{2p-1}a = q_{2p}, \qquad q_{2p}a = q_{2p+1}, \qquad q_{2p+1}a = q_3.$$

On $q_{2p+2}, \dots, q_{3p-1}$ it swaps successive pairs starting at the even index $2p+2$; an unmatched last state is fixed. Letter b swaps q_{2j} and q_{2j+1} for $0 \le j \le p-1$, has $q_{2p}b = q_2$, and swaps successive pairs on $q_{2p+1}, \dots, q_{3p-1}$ starting at $2p+1$, again fixing an unmatched last state. In particular, $q_0 b = q_1$ and $q_1 b = q_0$. The boundary assignments are compared with the published definition below.

For the proof, we write

$$M_i = q_i \ (0 \le i \le 2p), \qquad O_h = q_{4p-h} \ (p+1 \le h \le 2p-1).$$

The outer coordinate runs in the reverse direction. Away from the exceptions, on either branch, the letter a increases an odd coordinate by one and decreases an even coordinate by one; the letter b acts in the opposite direction. At the outer bottom $h = p+1$, the decreasing letter fixes the state. It is a when $p+1$ is even and b when $p+1$ is odd. The remaining exceptional rules are

$$M_0 a = M_0, \quad M_{2p}a = O_{2p-1}, \quad M_{2p}b = M_2, \quad O_{2p-1}a = M_3. \tag{1}$$

Figure 1 illustrates the two branches and their exceptional connections.

Both letters have rank $3p-1$. Indeed, their only double fibers are

$$a^{-1}(M_3) = \{M_4, O_{2p-1}\} = K_a, \qquad b^{-1}(M_2) = \{M_3, M_{2p}\} = K_b. \tag{2}$$

Each letter misses one image state: a misses M_{2p-1} and b misses M_{2p}. All other fibers are singletons. The underlying directed graph is strongly connected. The main matchings connect $M_0, \dots, M_{2p-1}$ in both directions, the final main a-edge reaches M_{2p}, and a then enters the outer branch. Its matchings connect every outer state in both directions, and its top returns to M_3; $M_{2p}b = M_2$ also returns to the main chain.

Boundary completion and scope of Conjecture 8

All values assigned by the piecewise formulas of [4], after substituting $k = p-2$, agree with the transitions above. For each p, those formulas leave exactly two transitions unspecified: $q_0 b$ and one action at q_{3p-1}. Table 1 records their completion from Figure 3.

The opposite terminal letter is already specified as a fixed point in the original formulas: $q_{3p-1}a = q_{3p-1}$ for odd p, and $q_{3p-1}b = q_{3p-1}$ for even p. Thus Table 1 completes the terminal matching and the $q_0 \leftrightarrow q_1$ b-edge in the diagram.

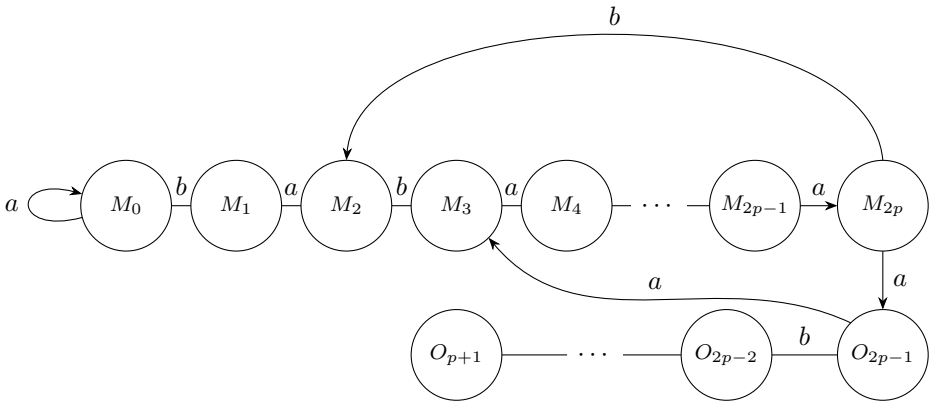


Figure 1: The transition structure of $\mathcal{A}_p$ (adapted from Figure 3 of [4], published under CC BY), shown schematically for $p \geq 5$. A labelled line without arrowheads represents a two-way transposition; a directed arrow represents a one-way transition. Unlabelled segments continue the alternating matchings. The outer bottom O_{p+1} is fixed by its decreasing letter. For $p = 4$, the outer branch is O_7, O_6, O_5 with no intermediate vertex; for $p = 3$, it is O_5, O_4 and the displayed labels O_{2p-2} and O_{p+1} denote the same vertex. The diagram is schematic.

State and letter	Parameter	Added transition
$q_0 b$	all $p \geq 3$	$q_0 b = q_1$
$q_{3p-1} b$	p odd	$q_{3p-1} b = q_{3p-2}$
$q_{3p-1} a$	p even	$q_{3p-1} a = q_{3p-2}$

Table 1: Boundary transitions drawn in Figure 3 of [4] but omitted from its piecewise formulas. For each fixed p, one of the last two rows applies.

Conjecture 8 of [4] prints $(ab)^p a (ab)^p$ as a shortest collision word and assigns it length $4n/3 = 4p$. The printed word has $4p + 1$ letters. Theorem 2 proves the numerical threshold for the completed automaton, using the shorter word $(ba)^p (ab)^p$. Proposition 3 proves synchronization for the same automaton.

3 Exact pair distances and the threshold

The unordered pair graph has vertex set $\binom{Q}{2}$. It has a directed edge $P \xrightarrow{x} Px$ for $x \in \{a, b\}$ whenever $|Px| = 2$; collapsing edges are omitted. Set

$$S_p = \{\{M_0, q\} : q \in Q \setminus \{M_0\}\}.$$

Distance from S_p means the minimum over its vertices. Call $P \xrightarrow{x} P'$ an *entrance* to a pair set X if $P \notin X$ and $P' \in X$. The nonstandard inverse fibers used in the entrance calculations are

Letter	Target	Full preimage
a	M_0	$\{M_0\}$
a	M_3	$\{M_4, O_{2p-1}\}$
a	M_{2p-1}	$\varnothing$
a	M_{2p}	$\{M_{2p-1}\}$
a	O_{2p-1}	$\{M_{2p}\}$
b	M_0	$\{M_1\}$
b	M_1	$\{M_0\}$
b	M_2	$\{M_3, M_{2p}\}$
b	M_{2p}	$\varnothing$

The remaining inverse fibers follow from the two-way matchings, with an unmatched terminal state fixed by its unmatched letter. A first collision must use a kernel pair, so

$$\mu(q_0) = 1 + \min\{\mathrm{dist}(S_p, K_a), \mathrm{dist}(S_p, K_b)\}. \tag{3}$$

Lemma 1. *For every $p \geq 3$,*

$$\mathrm{dist}(S_p, K_b) = 4p - 1, \qquad \mathrm{dist}(S_p, K_a) = 4p.$$

Proof. We prove the lower bounds first. Fix an arbitrary path $P_0 \xrightarrow{x_1} P_1 \xrightarrow{x_2} \cdots \xrightarrow{x_\ell} P_\ell$ in the distinct-pair graph, where $P_0 \in S_p$ and $x_t \in \{a, b\}$. All time bounds refer to its vertex P_t after t original letters. We first obtain lower bounds and then give matching paths.

1. The index bound and excluded region. Every original state index increases by at most one in one letter. Thus the trajectory starting at q_0 can have index no greater than the elapsed time t. Since one member of P_t is this trajectory, we obtain

$$t \geq \min\{i : q_i \in P_t\}. \tag{4}$$

In particular, for a mixed pair $P_t = \{M_x, O_h\}$ the original indices are x and $4p - h > 2p \geq x$, respectively. Thus (4) implies $t \geq x$ directly, without converting the outer coordinate back at every occurrence.

Let J consist of opposite-parity main pairs whose coordinate sum exceeds $2p + 1$. The only reachable member of J is K_b. To see this, an ordinary matching step on an opposite-parity pair preserves its sum, and an ordinary step on a same-parity pair preserves that parity type. A pair involving the fixed M_0 cannot acquire a sum above $2p + 1$. The b-jump from M_{2p} lands at M_2; its other main image is at most M_{2p-1}. Hence the resulting main-pair coordinate sum is at most $2 + (2p - 1) = 2p + 1$, so it cannot enter J, which requires a strictly larger sum. The a-step from M_{2p} leaves the main branch.

The terminal edge $M_{2p-1}a = M_{2p}$ also requires attention. A target in J containing M_{2p} has the form $\{M_{2p}, M_j\}$ with odd $j \geq 3$. For $5 \leq j \leq 2p - 3$, its only a-preimage is $\{M_{2p-1}, M_{j+1}\}$, already in $J \setminus \{K_b\}$ since its sum is $2p + j > 2p + 1$. The remaining odd endpoint $j = 2p - 1$ has no a-preimage. For $j = 3$, the preimage using M_4 is likewise already in $J \setminus \{K_b\}$, while $\{M_{2p-1}, O_{2p-1}\}a = K_b$. An outer-to-main step lands at M_3, and an opposite-parity sum above $2p + 1$ requires its other image to be M_{2p}, exactly the case just treated. Two outer states cannot both enter the main branch in one letter. Finally, $K_b a = K_a$, whereas $|K_b b| = 1$. No initial pair lies in J, and no edge from outside $J \setminus \{K_b\}$ enters that set. This proves the invariant by induction. Call a pair *admissible* if it is outside $J \setminus \{K_b\}$.

2. The first entrance potential. Put

$$C = \{\{M_x, M_{2p+1-x}\} : 1 \le x \le p\}, \qquad \phi_C(\{M_x, M_{2p+1-x}\}) = 2p - 2 + |x - 3|.$$

No initial pair lies in C. Each internal edge changes the smaller coordinate by at most one, so the potential increases by at most one. The complete list of admissible entrances from outside C is

Letter	Source	Target	Source-time bound
a	$\{M_{2p-3}, O_{2p-1}\}$	$\{M_3, M_{2p-2}\}$	$2p-3$
b	$\{M_{2p-2}, M_{2p}\}$	$\{M_2, M_{2p-1}\}$	$2p-2$

Here and below source-time bounds concern arrival before the entering letter. Both follow from (4); after adding one they equal the target potential. Ordinary inverse matchings preserve the opposite-parity sum, so they give sources in C. The double fibers at M_3 under a and at M_2 under b yield the displayed entrances. The missing images M_{2p-1} under a and M_{2p} under b exclude the endpoint targets; the fixed state M_0 lies outside C. Thus induction on the path, including any exits and reentries, proves

$$P_t \in C \implies t \ge \phi_C(P_t). \tag{5}$$

In particular, applying this inequality at $\{M_1, M_{2p}\}$ yields $\operatorname{dist}(S_p, \{M_1, M_{2p}\}) \ge 2p$.

3. The reflected entrance potential. Consider the disjoint tracks

$$A = \{\{M_x, O_{2p+1-x}\} : 2 \le x \le p\}, \qquad E = \{\{M_x, O_x\} : p+1 \le x \le 2p-1\}.$$

For $H = A \cup E$ set $\phi_H = 2p - 1 + x$, where x is the main coordinate. No initial pair lies in H. Ordinary internal edges change x by one. At the outer bottom, its decreasing letter exchanges A at $x = p$ with E at $x = p+1$; this also changes the potential by one. All admissible entrances from outside H are

Letter	Parameter	Source	Target	Source-time bound
a	all p	$\{M_1, M_{2p}\}$	$\{M_2, O_{2p-1}\}$	$2p$
b	all p	$\{M_{2p}, O_{2p-2}\}$	$\{M_2, O_{2p-1}\}$	$2p$
a	$p \ge 4$	$\{O_{2p-1}, O_{2p-3}\}$	$\{M_3, O_{2p-2}\}$	$2p+1$
a	$p = 3$	$\{O_5, O_4\}$	$\{M_3, O_4\}$	7

The first bound is (5); the others follow from (4). Adding one gives ϕ_H at each target.

The entrance table follows from the inverse fibers and the matching structure. In the interior of A, the coordinates have opposite parity, so both forward and inverse matchings preserve their sum. In the interior of E, the equal coordinates move together. At the outer bottom O_{p+1}, the state fixed by the decreasing letter allows the paired main-state coordinate to pass between the endpoints of A and E. At the outer top, $a^{-1}(O_{2p-1}) = \{M_{2p}\}$: target main coordinate 2 gives the first entrance, whereas the top target in E has main coordinate $2p-1$ and no a-preimage. Under b, the double fiber at M_2 gives the second entrance; its other source stays in H. To enter from two outer states under a, the main image must be M_3, hence the target is the $x = 3$ member

of A. Its ordinary outer preimage is O_{2p-3} for $p \geq 4$. For $p = 3$, $h = 4$ is the bottom and is fixed by a, so that preimage is instead O_4. These are the last two rows. The missing main images and the remaining ordinary inverse matchings give no further entrances. Therefore

$$P_t = \{M_x, O_h\} \in H \implies t \geq \phi_H(P_t) = 2p - 1 + x. \tag{6}$$

In particular, at $x = 2p-1$ this gives $\operatorname{dist}(S_p, \{M_{2p-1}, O_{2p-1}\}) \geq 4p-2$. For $p = 3$, the exceptional entrance reaches the member of A with $x = 3$, whose potential is $2p - 1 + 3 = 8$. Likewise, the top of E has $x = 2p - 1 = 5$, giving the lower bound $2p - 1 + 5 = 10$.

4. Entrance into the kernel regions. There is only one admissible incoming edge to K_b:

$$\{M_{2p-1}, O_{2p-1}\} \xrightarrow{a} K_b. \tag{7}$$

Indeed, b misses M_{2p}, and the other a-preimage is $\{M_4, M_{2p-1}\}$, in $J \setminus \{K_b\}$. Equation (6) now yields $\operatorname{dist}(S_p, K_b) \geq 4p - 1$.

To control K_a, we define

$$B = \{\{M_x, O_{2p+3-x}\} : 4 \leq x \leq p + 2\}, \qquad D = \{\{M_{h+2}, O_h\} : p + 1 \leq h \leq 2p - 2\}.$$

These tracks are disjoint. Their union U contains K_a and no initial pair. Its only admissible entrance from outside U is the a-transition from K_b to K_a. In fact, ordinary inverse matchings preserve the sum $2p+3$ in B, or the difference 2 in D. At the outer bottom the decreasing letter joins B at main coordinate $p + 2$ to D at main coordinate $p + 3$. The only U-target containing the outer top is the $x = 4$ member of B, namely K_a, whose a-preimage is K_b. A b-jump lands at M_2, and an outer-to-main a-jump at M_3, whereas every main coordinate in U is at least 4. Targets with main coordinate $2p$ have no b-preimage; those with main coordinate $2p - 1$ have no a-preimage. The other top endpoint of D, namely $\{M_{2p}, O_{2p-2}\}$, has its ordinary a-preimage in D for $p \geq 4$, and in B for $p = 3$. For $p = 3$, $D = \{\{M_6, O_4\}\}$, whose a-preimage is the $x = 5$ member of B because O_4 is fixed by a.

Every path reaching U has a first entrance immediately preceded by a visit to K_b. Hence its first entrance has length at least $4p$, and later visits cannot occur earlier. This proves $\operatorname{dist}(S_p, K_a) \geq 4p$. A final b-collapse costs at least $4p$, and a final a-collapse at least $4p + 1$.

5. Matching paths and the upper word. Put $c = ba$ and $u = ab$. Direct substitution in the transition rules gives cycles, in the displayed order,

$$\begin{aligned} R &= (M_0, M_2, M_4, \ldots, M_{2p}, M_1) \quad \text{under } c, \\ T &= (M_0, M_1, M_3, \ldots, M_{2p-1}, M_2) \quad \text{under } u. \end{aligned} \tag{8}$$

Both have length $p + 2$. In particular, $\{M_0, M_2\}c^p = \{M_{2p}, M_1\}$ and $M_1 u^p = M_2$. The u-chain starting at M_{2p} visits the even-indexed outer states upwards, reflects at the last original state, visits the odd-indexed outer states downwards, and then reaches M_2. It consists of M_{2p} and all $p - 1$ outer states, so its entrance length is p. Consequently

$$\{q_0, q_2\}(ba)^p(ab)^p = \{q_2\}. \tag{9}$$

Immediately before the last a in this word, the pair is $\{M_{2p-1}, O_{2p-1}\}$. After that a it is K_b, and the final b collapses it. These pairs are distinct until the last letter: an earlier collision would persist and contradict the distinct penultimate pair. Thus the prefix of length $4p - 1$ reaches K_b in the pair graph. Appending a, instead of the collapsing b, reaches K_a at length $4p$. This proves both upper distance bounds. □

Theorem 2. *For every $p \geq 3$, the automaton $\mathcal{A}_p$ defined in Section 2, with $n = 3p$ states, satisfies*

$$\mu(q_0) = 4p = \frac{4n}{3}.$$

The word $(ba)^p(ab)^p$ is a shortest word compressing q_0 with another state; it merges q_0 and q_2 at q_2.

Proof. Combine Lemma 1 with (3) and (9). □

4 A synchronization construction

We first map Q onto the $(p+2)$-cycle R under $c = ba$, then prepare and successively shorten an interval of cycle positions. Finally, u^p, for $u = ab$, merges the two remaining states. Identify the positions of R in (8) via the bijection

$$\rho : \{0, \ldots, p+1\} \longrightarrow R, \qquad \rho(0) = M_0, \quad \rho(i) = M_{2i} \ (1 \leq i \leq p), \quad \rho(p+1) = M_1.$$

For a set I of positions, write $\rho(I) = \{\rho(i) : i \in I\}$. Thus c adds one modulo $r = p+2$ to positions: $\rho(i)c = \rho((i+1) \bmod r)$.

The remaining main odd states descend by two under c until reaching M_1. The outer c-chain visits odd original indices upwards, reflects at the terminal state, visits even original indices downwards, and then reaches M_3 and M_1. Its largest distance into R is p. Since c permutes R, this proves the exact image identity

$$Qc^p = R. \tag{10}$$

For $j \geq p$, the return word $d_j = ac^j$ maps R into itself. Define its induced map on positions by $f_j(i) = \rho^{-1}(\rho(i)d_j)$. For $0 \leq i \leq p+1$, we have

$$f_j(i) = (j-i) \bmod r \quad (0 \leq i \leq p-1), \qquad f_j(p) = f_j(p+1) = (j+1) \bmod r. \tag{11}$$

For $1 \leq i \leq p-1$, $M_{2i}a = M_{2i-1}$ reaches M_1 after $i-1$ applications of c, giving the first formula. Position 0 is fixed by a. From position p, the letter a reaches the outer top, which enters R at M_1 after p applications of c. Position $p+1$ goes under a to M_2, giving the same residue. This verifies every position in (11).

For the preparation stage, we take

$$v_p = c^p u^{p-1} a^4 c^{p-2}, \qquad |v_p| = 6p-2.$$

For $1 \leq i \leq p-1$ the even states satisfy $M_{2i}u^{i-1} = M_2$. The u-cycle in (8) and the outer u-chain also give

$$M_0u^{p-1} = M_2u^p, \qquad M_1u^{p-1} = M_2u^{p+1}, \qquad M_{2p}u^{p-1} = O_{2p-1}.$$

Thus the images of $M_0, M_1, M_2, M_4, \ldots, M_{2p-2}$ under u^{p-1} are the $p+1$ distinct cycle states $M_2u, \ldots, M_2u^{p+1}$; these are precisely the states of T other than M_2. Consequently,

$$Ru^{p-1} = \{M_0, M_1, M_3, \ldots, M_{2p-1}, O_{2p-1}\}.$$

Under a^4, the two states M_{2p-1} and O_{2p-1} both reach M_4, while all remaining displayed states are fixed. The resulting image is $\{M_0, M_1, M_3, \ldots, M_{2p-3}, M_4\}$. Applying c^{p-2} gives, in R-positions,

$$Qv_p = \rho(I_{p-2}), \qquad I_t = \{0, \ldots, t\} \cup \{p, p+1\} \ (0 \leq t \leq p-2), \quad I_{-1} = \{p, p+1\}. \tag{12}$$

Indeed, the odd states yield positions $p-4, p-5, \ldots, 0, p+1$ when $p \geq 4$, while M_0, M_1, M_4 give $p-2, p-3, p$. For $p=3$, the prepared set before the final c is $\{M_0, M_1, M_3, M_4\}$; its image has positions $\{0,1,3,4\} = I_1$. This yields I_1 directly for $p=3$.

For $0 \leq t \leq p-2$, equation (11) gives

$$f_{p+t}(I_t) = I_{t-1}. \tag{13}$$

For $2 \leq t \leq p-2$, the interval $\{0, \ldots, t\}$ maps to the position set

$$\{(p+t-i) \bmod r : 0 \leq i \leq t\} = \{p,\, p+1,\, 0, \ldots, t-2\}, \qquad r = p+2.$$

The pair $p, p+1$ merges to residue $(p+t+1) \bmod r = t-1$. Combining the two images gives I_{t-1}: the positions $0, \ldots, t-1$ together with p and $p+1$. At $t=1$, the interval maps to $\{p, p+1\}$ and the merged image is 0, giving I_0. At $t=0$, the interval maps to $\{p\}$ and the merged image is $-1 \bmod r = p+1$, giving $I_{-1} = \{p, p+1\}$.

Proposition 3. *For every $p \geq 3$, a reset word for $\mathcal{A}_p$ is*

$$w_p = c^p u^{p-1} a^4 c^{p-2} (a c^{2p-2})(a c^{2p-3}) \cdots (a c^p) u^p. \tag{14}$$

It maps Q to $\{q_2\}$ and has length $3p^2+4p-1$. In particular, $\mathcal{A}_p$ is synchronizing and

$$\mathrm{rt}(\mathcal{A}_p) \leq 3p^2 + 4p - 1.$$

Proof. Equation (12) initializes the image. Applying (13) in descending order $t = p-2, \ldots, 0$ leaves positions $p, p+1$, that is $\{M_{2p}, M_1\}$. The final u^p maps both to M_2. Counting every a, b letter gives

$$|w_p| = (6p-2) + \sum_{j=p}^{2p-2} (2j+1) + 2p = (6p-2) + (3p-1)(p-1) + 2p = 3p^2 + 4p - 1. \qquad \square$$

Corollary 4. *If $\mu(q) \leq \alpha n + o(n)$ uniformly over all binary strongly connected synchronizing n-state automata whose letters have corank one, and all their states q, then $\alpha \geq 4/3$.*

Proof. For the automata $\mathcal{A}_p$, the uniform hypothesis gives $4p \leq 3p\alpha + o(p)$. Dividing by $3p$ and letting $p \to \infty$ proves $\alpha \geq 4/3$. $\square$

5 Computational checks and reproducibility

The checkers and source files are available at `https://github.com/dongxuelian2/cerny-direct-attack` (in `papers/figure3/` and `scripts/direct_attack/`). The earlier verification programs can be inspected at the permanent revision `e8e9cf4f`. The checker `check_figure3_manuscript.py` reconstructs the transitions and verifies the kernel distances and reset-image identities for $3 \leq p \leq 128$. Separate certificate tests cover $p = 3, \ldots, 64$ and $p = 100, 128, 256, 512, 1024$.

An exhaustive breadth-first search of reachable subsets, implemented in `check_figure3_exact_rt.py`, computes exact reset thresholds for $p = 3, \ldots, 10$:

p	3	4	5	6	7	8	9	10
$\mathrm{rt}(\mathcal{A}_p)$	38	63	94	131	174	223	278	339

These values equal $3p^2+4p-1$ throughout the tested range. The same checker compares every transition assigned by the original piecewise formulas with the completed family. The computations verify finite cases; the all-parameter bounds are proved in Sections 3 and 4.

6 Conclusion

The family $\mathcal{A}_p$ attains the specified-state collision threshold $4n/3$ and admits a reset word of length $3p^2 + 4p - 1$, asymptotic to $n^2/3$ and well below the Černý bound $(n-1)^2$. Whether a uniform linear bound for $\mu(q)$ holds in the strongly connected case remains open.

Three questions arise: Is $4/3$ optimal as a lower-bound coefficient within the binary strongly connected class? Does the reset-word construction attain $\mathrm{rt}(\mathcal{A}_p)$ for every p, as it does for $3 \le p \le 10$? Can similar pair-graph entrance bounds be established for other families?

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